Treating randomized experiment data with attention to details

Goal: Using available experimental data, implement a policy assigning 0 / 10 / 20 / 30 / 40% vouchers in an optimal way across eligible users to maximise incremental order value, while keeping total expected voucher spend within 3,500 CU for upcoming weeks

5 × 2	voucher × free delivery treatment groups, fully randomised
~60k	eligible users per week, stable population
6	weeks of history (9–14), re-randomised weekly
1	treatment per user-week unit, treatment arms equal in size

What was checked

Assignment was indeed completely random: A propensity model can't predict the voucher arm from features (ROC-AUC ~ 0.50), nor free delivery at inference (ROC-AUC ~ 0.49)

No carry-over: Last week's voucher doesn't move this week's conversion ($p = 0.46$), so moving from panel design to flat with one-week decisions is fine

Outflow doesn't seem to be treatment-driven: ~4% of users leave each week at the same rate in every arm, but unexplained by existing features

Population is stable: Feature distributions hold week to week; cities behave alike

What we had to be careful about

Week 16 should not really be scored. Its features already reflect week 15, as if that week's decisions had already been made, which breaks causality for our interventions and doesn't match production logic, so we assign interventions for week 15 only

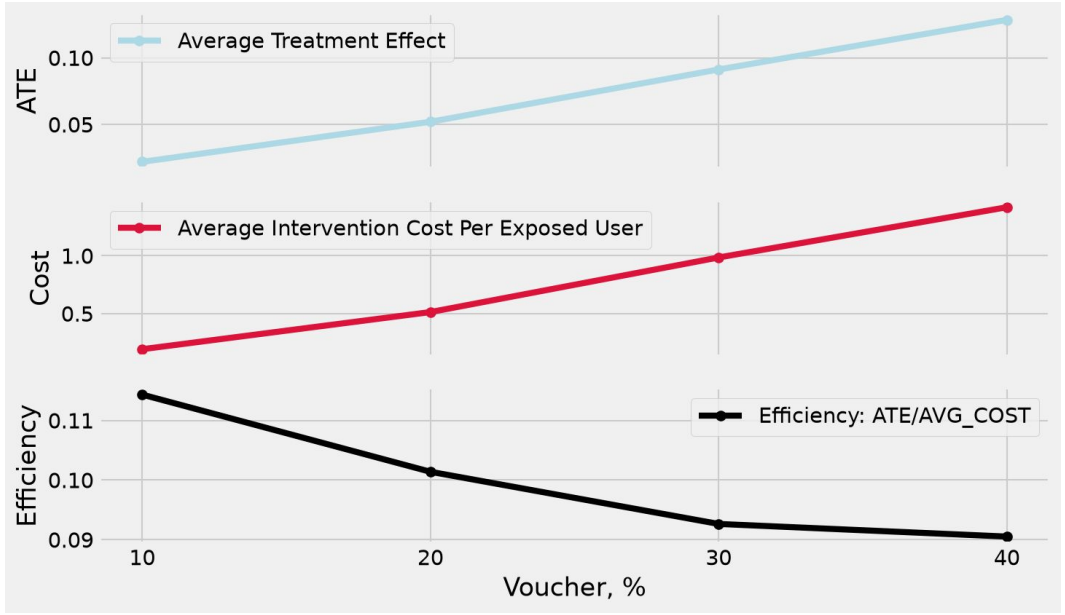
Budget is ambiguous: Per week or total isn't clarified, and it's ~10× below historical weekly voucher cost. We use 1,750 CU / week

Binary target: "Ordered" is a binary flag, no explicit order value and no order count post-treatment are available within observation window

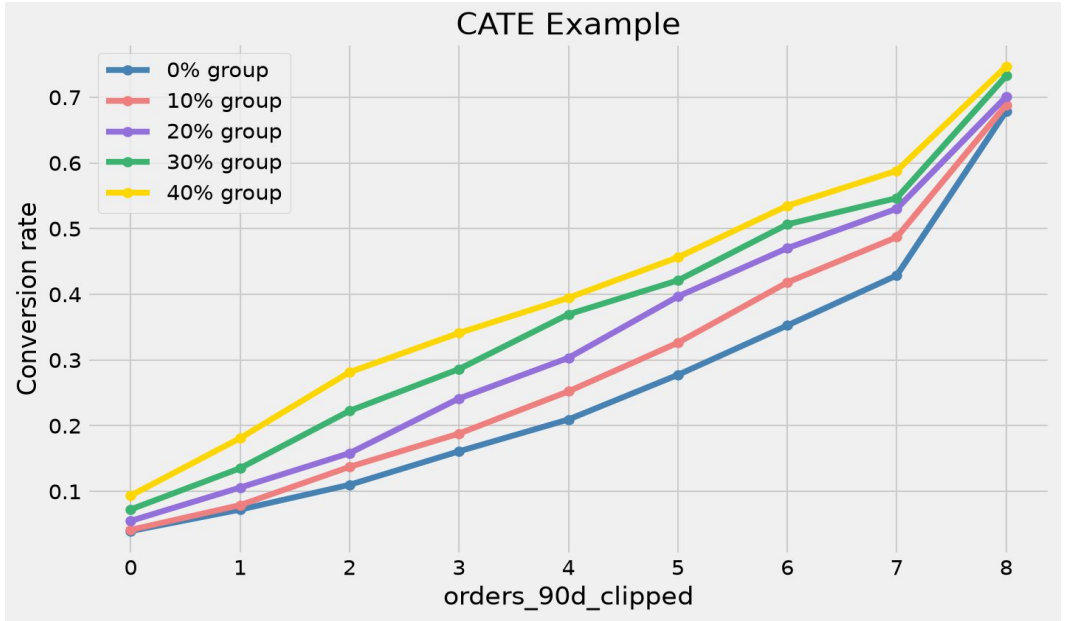
Free delivery is fixed going forward: It goes into the models as a covariate impacting outcome

Higher discounts convert more, but it doesn't come free
Treatment effect heterogeneity is present

Average effect, cost and efficiency by voucher group



Conversion by past orders for each voucher group



Uplift per CU spent falls with discount increase

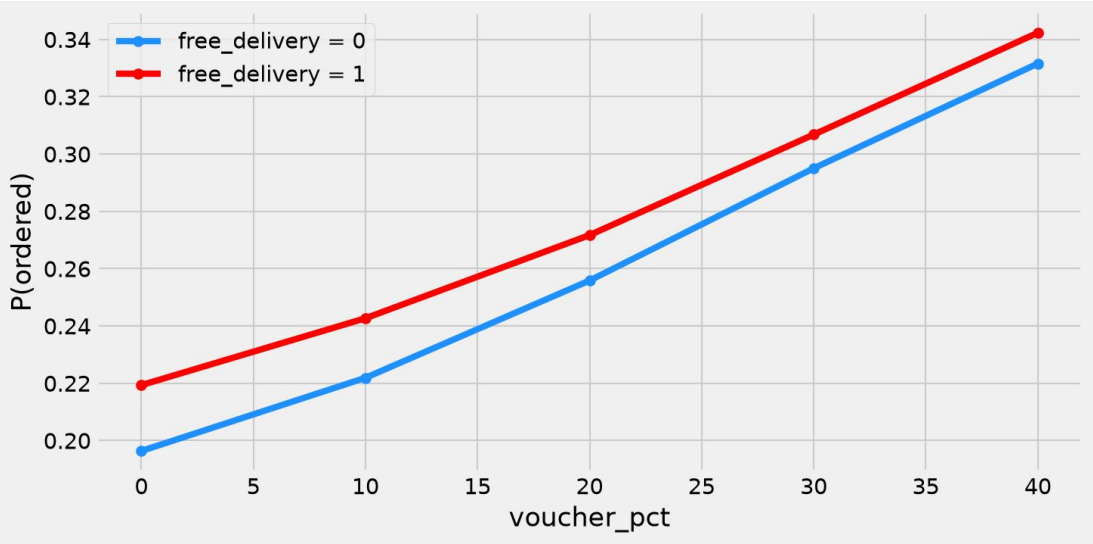
Uplift grows from +2.2 pp at 10% to +12.8 pp at 40%, but cost per user grows faster: efficiency 0.114 → 0.091. We might think about getting monetary value back, not only increasing order conversion - this might be different objectives.

Treatment effect differs by users

At 40%, uplift is +5 pp for users with no orders in 90 days and +19 pp for users with 4 as one example - pointing we might be able to model CATE, not just ATE.

Randomized Experiment Data: Free Delivery Interaction & ATE Dynamics

FD increases conversion at every voucher level, but less at 30–40%



Weekly order conversion rate by voucher arm, weeks 10–14



Discounts and Free Delivery are partial substitutes

Joint Wald test on the voucher × FD interactions: p = 0.059. FD adds +2.3 pp with no voucher, curves run parallel before 20% and start to converge from there. FD is kept as a model covariate.

Conversion levels slightly drift, but not the ATE

Weeks 11–12 are 2.7–2.9 pp lower than week 10, week 14 is 1.9 pp higher again, but the ATE mostly holds and arms order is preserved every week,

3 different learners, consistent and proper evaluation

META-LEARNER

S-learner (LightGBM)

- One outcome model P(order | X, d, fd) with voucher level as a feature
- Uplift = P(order | X, d, fd) − P(order | X, 0, fd) for each treatment d
- Simple, any classic ML-classifier could be used, but the effect is a by-product of main target prediction, conditional uplift identification might be weak in practice

TREE-BASED MODELS CAUSAL BY DESIGN

Uplift Tree & Uplift Random Forest

- Splits maximise treatment-vs-control conversion target difference (e.g. using KL criterion) conditioned on feature value buckets
- One model covers all five voucher levels
- URF averages multiple deep trees for stability and variance reduction

Validation Protocol

	weeks 10–13	week 14
80% users	train	out-of-time
20% users	test	holdout

Split by user, so train and test don't overlap by id. Week 14 checks drift in time and used for policy evaluation as full population slice. Holdout is unseen users in an unseen week. Each voucher level's modelled uplift is evaluated against control on 1-vs-1 basis.

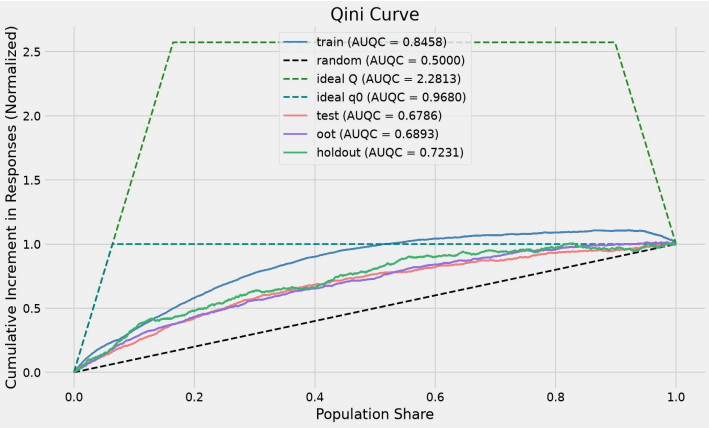
Qini T vs C, macro-averaged over 4 non-control voucher levels

Sample	S-Learner	Uplift Tree	Uplift RF
Train	0.083	0.130	0.205
Test	0.059	0.036	0.066
Out-of-time	0.055	0.041	0.054
Holdout	0.049	0.049	**0.062**

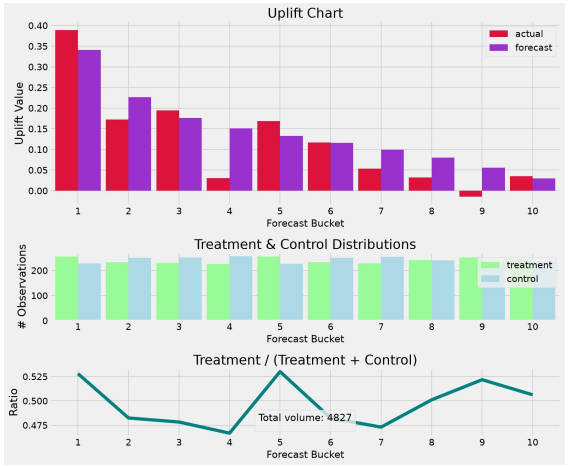
URF wins consistently. Train scores are over-positive, we choose based on unseen out-of-sample data, both in time and out-of-time.

Uplift Random Forest ranks users well enough and stays calibrated

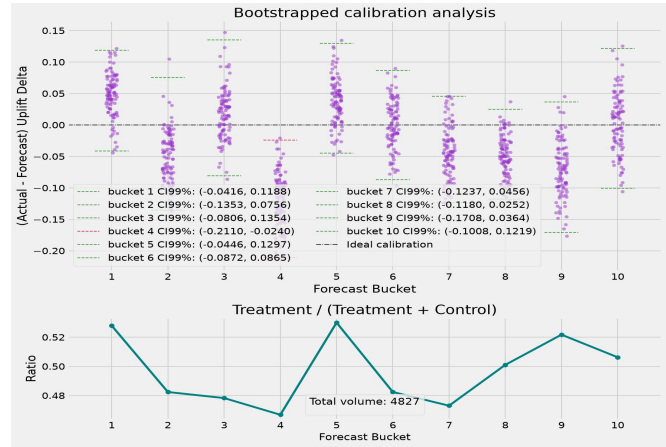
Qini (40% vs control): ranking holds across sets



Actual vs Empirical Uplift Match By Buckets

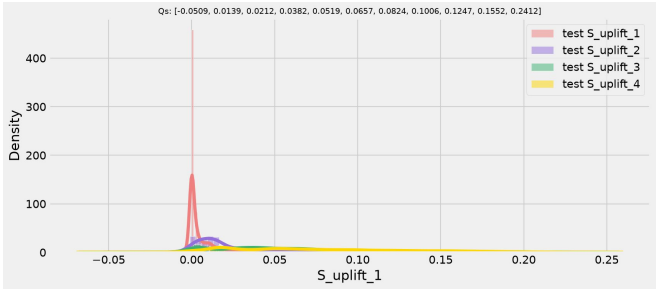


Calibration holds: fact within CI on holdout

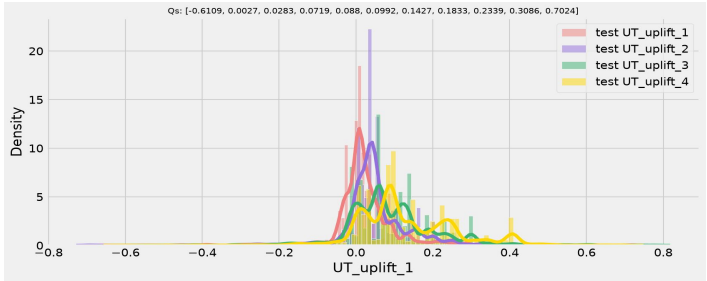


How different learners' uplift predictions are

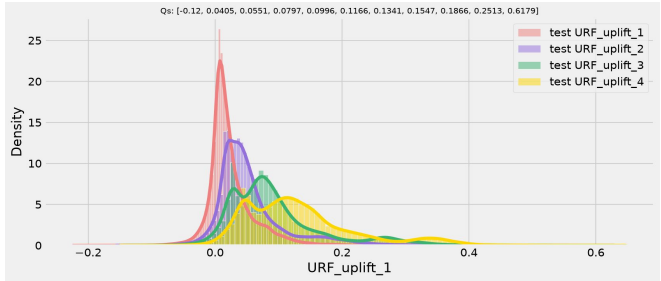
S-learner: squeezed near 0, treatment effect is dampened by other features



Uplift Tree: Estimates are discrete and noisy



URF: Distribution is smoother, estimates are stable



Target for monetary incrementality, control for actual spend

1 USER-BASED VALUE

$AOV(X) \times \Delta Conversion(d, X)$

Used aov_90d, or a forecast from historical non-purchase based features for the not recently purchasing customers

2 SPEND

$P(voucher_used \mid d, X) \times voucher_cost$

Utilization is modelled on user level (ROC-AUC 0.78), cost capped at 9 CU as per business setup

3 ASSIGN

LP: max Σ value, Σ spend ≤ B

At most one voucher per user, solved with cvxpy

Spend model iterations on week 14 (budget 1,750 CU)

	Expected spend per user formulation	Vouchers assigned	Realised spend	Realised / Expected	Value / CU
v1	P(order|X) · P(voucher_used | order, d) · E(voucher_cost|voucher_used, d)	8,371	3,607	2.06×	3.8 vs 3.2
v2	P(voucher_used|d) · min(d · AOV(X), 9)	7,107	5,368	3.07×	4.3 vs 2.7
v3	P(used | X, d) · min(d · AOV(X), 9)	5,169	2,237	**1.28×**	**5.2 vs 3.2**

Realised: IPW spend of selected users (plan 1,750). Value / CU: order value gained per CU, policy* vs random assignment

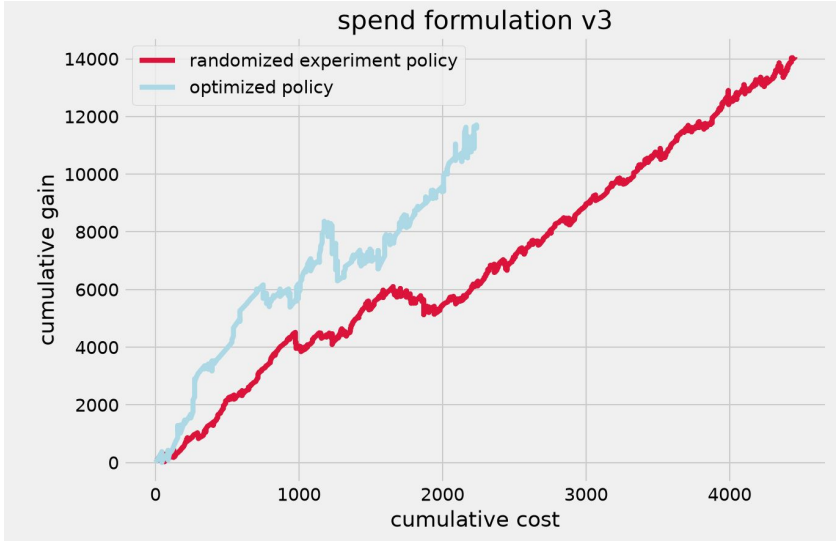
Off-Policy Evaluation For Gain Estimation

Used off-policy evaluation of on week-14 experiment randomised data, using IPW with known propensity 0.2, measured against control no-voucher policy - for both experiment data and the optimization-based assignments.

Safety Factor For Better Spend Control

The users getting selected by our policy still use vouchers more often as per experimental data then our model captures on average, so even v3 overspends 1.28×. Week 15 is solved at 1,750 / 1.25 = 1,400 CU.

v3: cumulative value gain vs cumulative cost under policy-based ranking



Voucher assignment is not-uniform, mostly 10% discount, within budget

4,197

users are assigned a discount voucher

7.0% of 60,208 eligible users, rest are left “organic”

1,750 CU

expected voucher spend

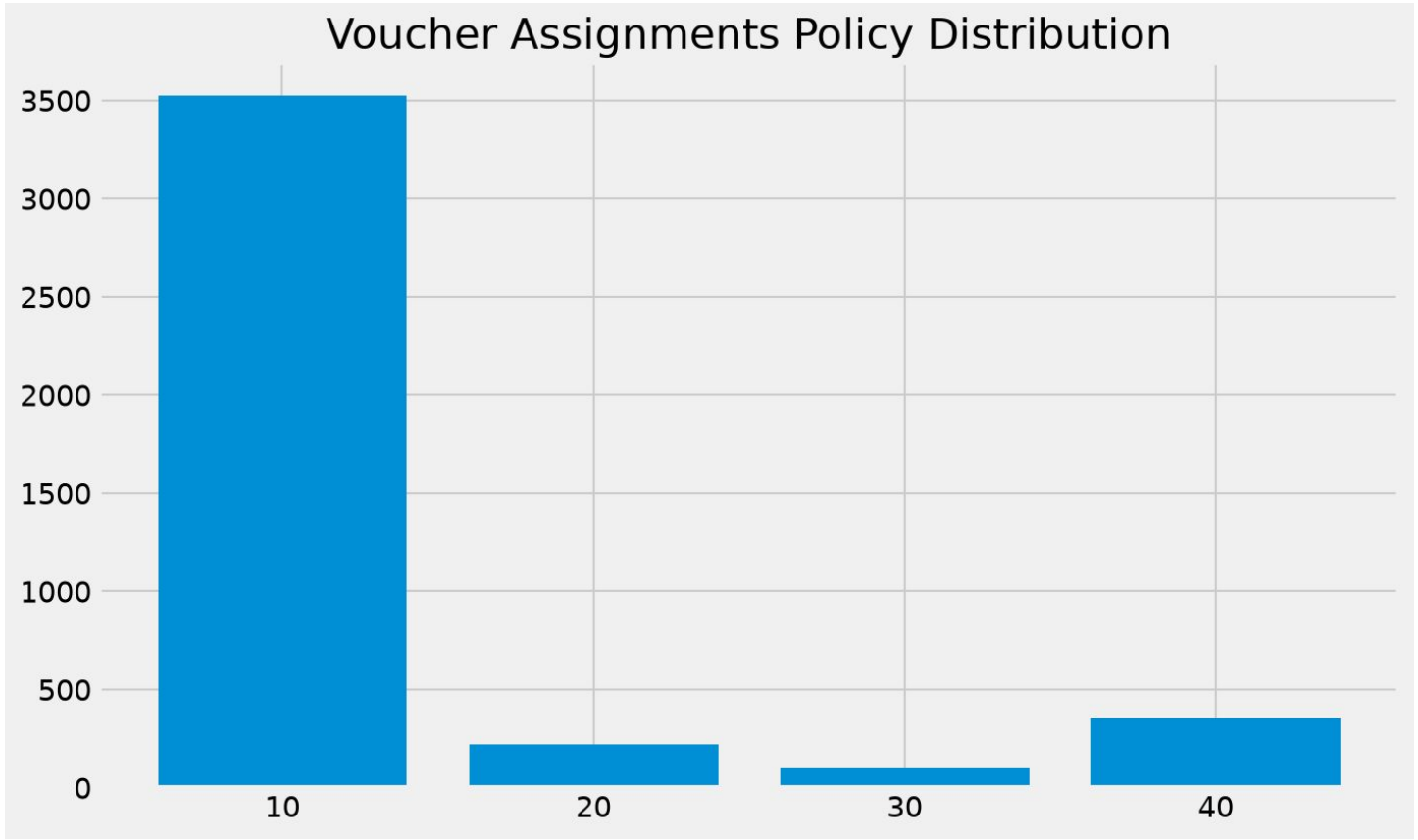
1,400 CU planned × 1.25 safety factor = full weekly budget

13,650 CU

expected incremental order value

Model estimate: URF uplift × AOV over assigned users

Voucher mix: 3,523 at 10% (84%), 220 at 20%, 100 at 30%, 354 at 40%



Can we make it better?

Current limitations

Historical AOV as a value proxy: We assume vouchers and FD don't change order value or orders per converting customer, there's no post-treatment data to check

Binary target: Extra orders from users who would have ordered anyway are not known explicitly

Production FD logic is unknown: If FD assignment follows a rule driven by unobserved confounders, uplift estimates would drift

Spend control is not bulletproof: The conservative factor comes from one week estimations, variance is likely large

Next steps planning

Proper control for lookahead: Inference data one week at a time, so features are actualized and don't account for not yet made decisions or increase the gap between assignment optimizations

Track more outcomes after treatment: Order value and order count per user-week, to optimise for value (margin and/or repeated orders) directly

Clarify business scenarios: Why users drop out of the segment, how FD is assigned in production, should we cover more user segments or shrink the coverage

Horizon increase: A multi-week design without follow-up vouchers to see longer effects on retention

Advanced tooling: explore other meta-learners (T-, R-, DR-,) with automated tuning and model selection + bootstrapped estimates for off-policy evaluations

Next Experiment Design Proposal: validate the policy in a controlled experiment before scaling it

Design

Policy vs **Random 10-20-30-40** vs **No-Voucher Control**

Split groups w.r.t business risk-appetite while controlling for MDE and FPR

Reevaluate gain/cost tradeoff, compare with offline evaluation

Primary metrics

- Incremental total order value
- Incremental user-conversion rate

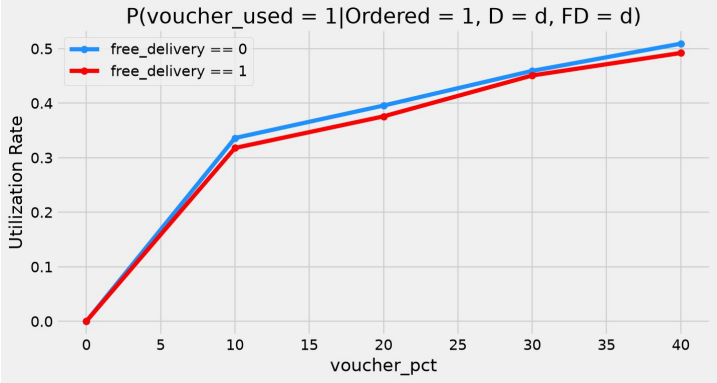
Guardrails

- Realised voucher spend vs plan
- Utilization rate
- Average voucher cost
- Comms delivery funnel rates

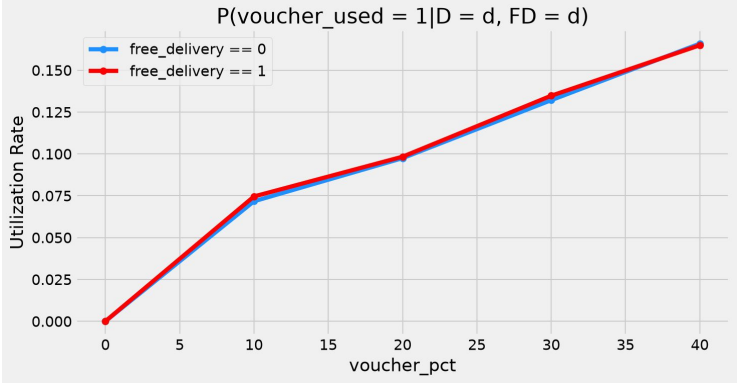
APPENDIX A1

Optimization setup iterations: why naive cost estimates are dangerous

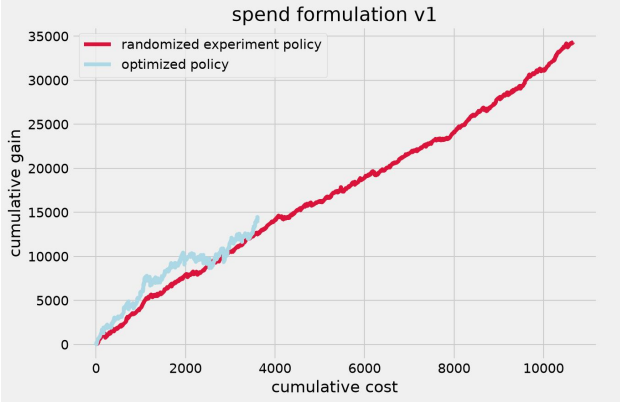
v1 input: P(voucher_used | ordered =1, d, FD), arm averages



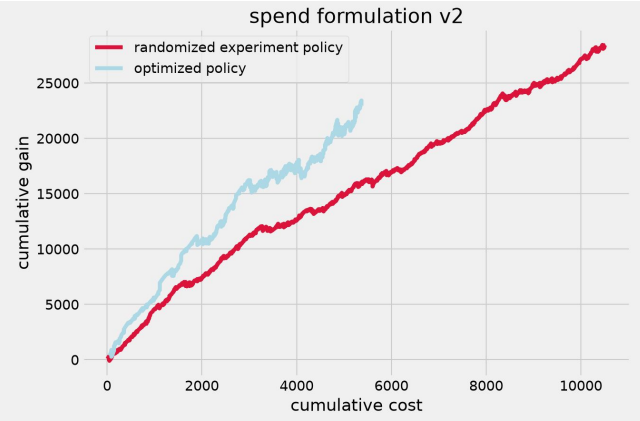
v2 input: P(voucher_used | d, FD), arm averages



v1 evaluation results: cumulative order value gain vs cumulative cost



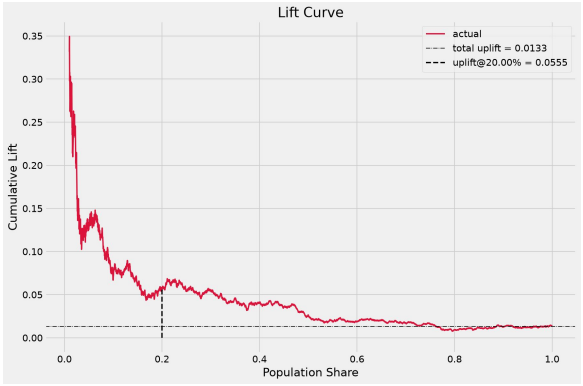
v2 evaluation results: cumulative order value gain vs cumulative cost



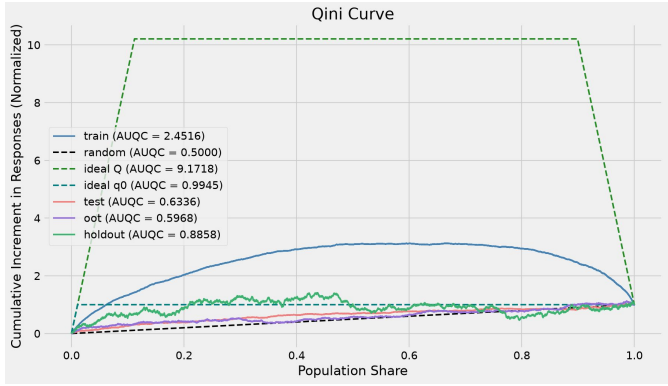
APPENDIX A2

Evaluation Techniques

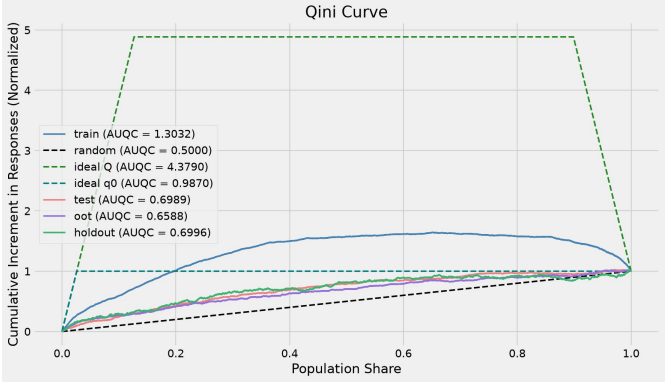
Lift, 10% vs control: top 20% under treatment get 5.6% vs 1.3% overall



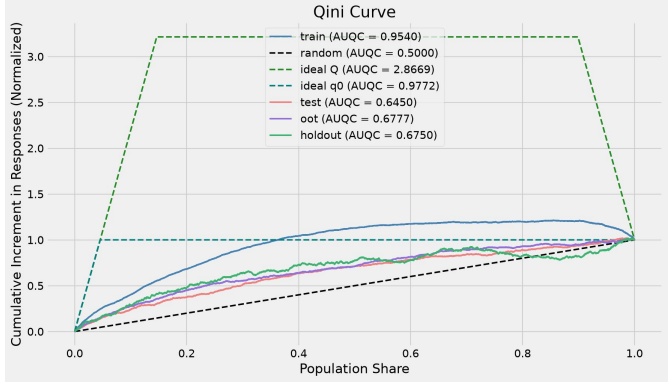
Qini, 10% vs control: the noisiest arm



Qini, 20% vs control



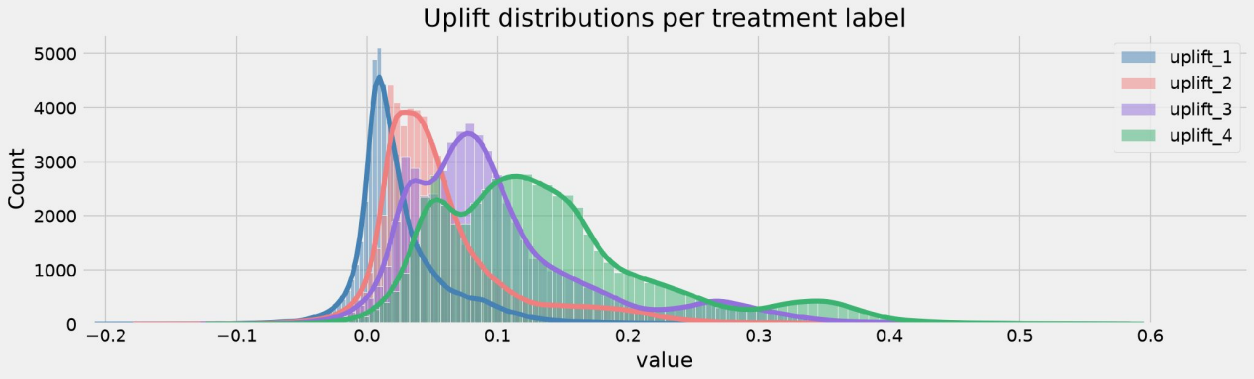
Qini, 30% vs control



APPENDIX A3

Uplift predictions distributions analysis

Uplift scores per discount group: larger vouchers both shift average effect and increase effect range



P(conversion) per discount group

